A Summary of “Multimodal Variational Auto-encoder based Audio-Visual Segmentation”

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The research paper “Multimodal Variational Auto-encoder based Audio-Visual Segmentation” was written by researchers Mao, Zhang, Xiang, Zhong, and Dai. In this paper, the researchers embark on the goal of audio-visual segmentation. Audio-visual segmentation is the process of separating the audio components of a video from the visual components in the video that produces the sound.

In the context of audio-visual segmentation, modality refers to the specific sense being evaluated by the model. For the audio component, the model evaluates the sound and for the visual component, the model evaluates the displayed video frames. The researchers highlighted the fact that each modality contains both shared and specific information. The audio modality contains information from the object producing the sound as well as any background noise. For the visual modality, while the object producing the sound may be displayed, it is typically only a portion of the entire scene. For example, in a video of a man playing a piano on stage in front of an audience, the audio component would certainly contain the sound of the piano, but it may also contain the noise of an audience member’s cell phone ringing. For the visual component, the piano would be the main sound producer, however, the video scene may also contain the musician as well as the stage. The goal of audio-visual segmentation involves creating a model able to evaluate the entire scene and identify the music as main sound and the piano as the sound producer while filtering out the background noise of the cell phone and objects in the video that are not producing the sound, such as the musician and stage.

The researchers acknowledge that while a foundational model for audio-visual segmentation is available, the existing state-of-the-art model provides reasonable performance but soft constraints. The soft constraints in the foundational model allows a pathway for the model to “regress the final segmentation maps by only taking video as input” (Mao, Zhang, Xiang, Zhong, & Dai, 2023). Due to the foundational model lacking strict sorting rules for the audio and visual components, there are no guidelines to prevent the model from overfitting. The researchers responsible for the foundational model acknowledge this limitation by demonstrating that even when the audio components are removed from the scene, the model can still determine the sound producers from just the video component with relatively high accuracy (Zhou, et al., 2022). This high accuracy is not from the model being able to correctly narrow down the sound producer but rather is due from the model memorizing objects that are more likely to be producing the sound in a scene.

In “Multimodal Variational Auto-encoder based Audio-Visual Segmentation”, the researchers propose a novel state-of-the-art model (Explicit Conditional Multimodal Variational Auto-Encoder, or ECMVAE) that performs audio-visual segmentation with strict guidelines that force the model to explicitly determine the contribution from each modality. It guarantees that the model cannot resort to memorizing common sound producers by implementing latent space factorization with orthogonality and shared information completeness.

Latent space factorization allows the model to understand enough about each component so that it can differentiate between the audio and visual inputs. The researchers use this differentiating ability to separate the audio and visual components into both modality-specific and modality-shared representations. Latent space factorization is analogous to the researchers separating scenes into three different buckets: a shared bucket for the sound producer in the scene, an audio specific bucket for the background noise, and a video specific bucket for the visual clutter in the scene. To ensure each representation, or “bucket”, captures unique data, the researchers apply an orthogonality constraint. This constraint uses the squared Frobenius norm to calculate the difference loss with a heavier loss applied if overlap is found between features in the different representations. Put simply, the orthogonality constraint penalizes the model for not setting clear boundaries between the sound producer representation and the background noise or visual clutter representations.

In regard to shared information completeness, the researchers combine the shared representations with each individual modality-specific representation. The combined representations are treated as “two different views of the same target” (Mao, et al, 2023). The researchers then applied mutual information maximization to obtain a similarity score between the combined representations.

In conclusion, the researchers observed that their segmentation method consistently outperformed the existing state-of-the-art model. Specifically, their semi-supervised Single Sound Source Segmentation (S4) scored 3.54 points higher Mean Intersection over Union (mIOU) than the existing model when using ResNet50 and 3.00 points higher mIOU when using PVTv2. Also, their S4 with ResNet50 model demonstrated a 0.017 higher F-score and their S4 model with PVTv2 resulted in a 0.022 higher F-score than the existing model. The researcher’s Multiple Sound Source Segmentation (MS3) model earned 0.81 higher mIOU when using ResNet50 and 3.84 higher mIOU when using PVTv2 with an increase in F-score of 0.029 and 0.063, respectively. The increases in mIOU and F-scores confirm Mao, Zhang, Xiang, Zhong, and Dai's model segmentation method indeed outperforms the existing state-of-the-art model.

References

Mao, Y., Zhang, J., Xiang, M., Zhong, Y., & Dai, Y. (2023). *Multimodal Variational Auto-encoder based Audio-Visual Segmentation*.

Zhou, J., Wang, J., Zhang, J., Sun, W., Zhang, J., Birchfield, S., . . . Zhong, Y. (2022). *Audio–Visual Segmentation*.